

Langchain : How do input variables work, in particular how is "context" replaced by what I want in the Template?

Asked 1 year, 8 months ago Modified 1 year, 7 months ago Viewed 8k times

I'm learning about langchain I had trouble understanding how templates work.

2

```
from langchain.prompts import PromptTemplate
prompt_template = """Use the following pieces of context to answer the question
at the end. If you don't know the answer, just say that you don't know, don't try
to make up an answer.
```

```
{context}
```

```
Question: {question}
Answer in Italian: ""
PROMPT = PromptTemplate(
    template=prompt_template, input_variables=["context", "question"]
)
```

Here there is a context input variable. What is this "context" ? I mean conceptually I know what it is, but how does it work ? Why does it corresponds exactly to some parts of the documents, since the function only use an embeddings database ? Why is it between {} ? Digging into the source code I also found a 'summaries' variable,

Thanks in advance :)

[python](#) [langchain](#) [Edit tags](#)

[Share](#) [Edit](#) [Follow](#) [Close](#) [Flag](#)

asked Oct 18, 2023 at 11:59

 [Morick](#)
21 1 2

1 Answer

Sorted by: [Reset to default](#) [Date modified \(newest first\)](#) 

context and question are placeholders that are set when the LLM agent is run with an input. This is why they are specified as input_variables when the PromptTemplate instance is created.

1

These placeholders are keys in the input dictionary fed to the langchain chain instance.

As an example, if I were to use the prompt template in the original post to create an instance of an LLMChain, I would specify context and question in the input to this LLMChain

```
from langchain.chains import LLMChain
from langchain.llms import OpenAI

chain = LLMChain(llm=OpenAI(temperature=0.9), prompt=PROMPT)

response = chain.run({
    "context": "Italy is home to some of the finest grapes in the world",
    "question": "What are the top three wine exports from Italy?"
})

print(response)
```

'Chianti, Prosecco e Barolo.'

[Share](#) [Edit](#) [Follow](#) [Flag](#)

answered Nov 15, 2023 at 5:42

 [curiouscat](#)
21 1

Comments 

 Add a comment

Start asking to get answers

Find the answer to your question by asking.

[Ask question](#)

Explore related questions

`python` `langchain`

See similar questions with these tags.